Commonness of independent apices of trees and even cycles

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Abstract

Let H^{+k} denote the graph obtained from H by adjoining an independent set of k vertices, each joined to every vertex of H. We prove that H^{+k} is common for every positive integer k when H is a tree or an even cycle. These results establish two infinite families of the independent-apex conjecture of Grzesik, Lee, Lidický, and Volec. For tree bases, we obtain a sharp density inequality by extending the law of a random book along the tree and estimating its relative entropy against the homomorphism weight. For even cycles, three exact six-vertex flag algebra certificates yield a weighted fourth-moment comparison. Spectral inequalities and convexity then extend this comparison to all even cycle lengths and all apex numbers. For at least three apices, we obtain a stronger comparison with the Ramsey multiplicity of the base cycle.

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1 Introduction

A graphon is a symmetric measurable function $W : [0,1]^2 \to [0,1]$; we interpret W and $1-W$ as the densities of the two colors. For a finite simple graph F with $e(F)$ edges, define

$$t(F,W) \coloneqq \int_{[0,1]^{V(F)}} \prod_{uv \in E(F)} W(x_u, x_v) \prod_{u \in V(F)} \mathrm{d}x_u, \qquad M(F,W) \coloneqq t(F,W) + t(F, 1-W),$$

so that $M(F,W)$ is the Ramsey multiplicity of F. The constant graphon $W \equiv 1/2$, a uniformly random coloring, is a natural baseline: it gives $M(F, 1/2) = 2^{1-e(F)}$. A graph F is *common* if no graphon does better, that is,

$$M(F,W) \geq 2^{1-e(F)} \qquad \text{for every graphon } W. \tag{1.1}$$

The triangle is common by Goodman's theorem [4], and Sidorenko extended this to every odd cycle [13]. Hatami, Hladký, Kráľ, Norine, and Razborov showed that the wheel with five spokes is common [6]. Kráľ, Volec, and Wei constructed common graphs of arbitrarily large chromatic number [10], and Ko and Lee made them, in addition, arbitrarily highly connected [8].

Erdős conjectured that every complete graph is common [3]. Thomason disproved this by showing that K_4 is uncommon [17], and Jagger, Šťovíček, and Thomason proved that every graph containing K_4 is uncommon [7].

For $k \geq 1$, write H^{+k} for the graph with vertex set $V(H) \sqcup \{z_1, \dots, z_k\}$ and edge set

$$E(H) \cup \{vz_i : v \in V(H),\ 1 \leq i \leq k\}.$$

The added vertices are pairwise nonadjacent; we call them independent apices. This operation connects commonness with Sidorenko's conjecture. A bipartite graph H has the *Sidorenko property* if

$$t(H,W) \ge t(K_2,W)^{e(H)} \qquad \text{for every graphon } W.$$

Sidorenko's conjecture asserts this property for every bipartite graph [14]. Applying it to both colors and using convexity immediately gives commonness of H. Sidorenko proved the stronger fact that H^{+1} is common whenever H is connected and has the Sidorenko property [15].

Grzesik, Lee, Lidický, and Volec proposed extending this result to arbitrarily many independent apices [5, Conjecture 1.3].

Conjecture 1.1 (Grzesik–Lee–Lidický–Volec)**.** *If H is a connected bipartite graph with the Sidorenko property, then H^{+k} is common for every positive integer k.*

Their paper verifies the conjecture for connected bipartite graphs on at most five vertices [5, Theorem 1.4]. It also treats the graphs $C_{2\ell}^{+2}$, called even beachballs, for all $\ell \ge 2$ [5, Theorem 4.3]. More recently, Kráľ, Krnc, and Lamaison proved that T^{+k} is common for every tree T when $1 \le k \le 5$ [9, Theorem 1]. We extend these results to all $k \ge 1$ for trees and even cycles, thereby establishing Conjecture 1.1 for two infinite families of base graphs.

1.1 Main results

Throughout, for a graph G we write $V(G)$ and $E(G)$ for its sets of vertices and edges, and $v(G) \coloneqq |V(G)|$ and $e(G) \coloneqq |E(G)|$ for their numbers. We use the convention $0^0 = 1$. The path with j edges is denoted by P_j; in particular, P_2 has three vertices.

Theorem 1.2 (Tree apices)**.** *For every tree T with at least one vertex and every integer $k \ge 1$, the graph T^{+k} is common. More precisely, if T has $n \ge 2$ vertices, then*

$$M(T^{+k},W) \ge \frac{M(K_3,W)^{k(n-1)}}{M(P_2,W)^{(k-1)(n-2)}} \tag{1.2}$$

for every graphon W.

The inequality (1.2) implies commonness when $n \ge 2$. Writing $\sigma \coloneqq M(P_2,W)$ and $\tau \coloneqq M(K_3,W)$, the right side of (1.2) equals $\tau^k\big[\tau(\tau/\sigma)^{k-1}\big]^{n-2}$. Goodman's identity $\tau = \frac{3}{2}\sigma - \frac{1}{2}$ [4], together with $\sigma \ge \frac{1}{2}$ (the commonness of P_2), gives $\tau \ge \frac{1}{4}$ and $\tau/\sigma \ge \frac{1}{2}$. Therefore

$$M(T^{+k},W) \ge \tau^k\big[\tau(\tau/\sigma)^{k-1}\big]^{n-2} \ge 2^{2-(k+1)n} = 2^{1-e(T^{+k})}.$$

Note that if T has one vertex, then T^{+k} is a star, whose commonness follows from the Sidorenko property.

We prove Theorem 1.2 by establishing the following one-color inequality with the entropy method.

Theorem 1.3. *For a tree T on $n \ge 2$ vertices, an integer $k \ge 1$, and a graphon W,*

$$t(T^{+k},W)\,t(K_2,W)^{n+k-3}\,t(P_2,W)^{(k-1)(n-2)} \ge t(K_3,W)^{k(n-1)}. \tag{1.3}$$

To see why Theorem 1.3 implies (1.2), given a graphon W, let $\overline{W}$ be the graphon obtained by placing rescaled copies of W and $1-W$ on $[0,\frac{1}{2}]^2$ and $(\frac{1}{2},1]^2$, with no edges between the two halves, that is,

$$\overline{W}(x,y) \coloneqq \begin{cases} W(2x,2y), & x \le \frac{1}{2} \text{ and } y \le \frac{1}{2}, \\ 1-W(2x-1,2y-1), & x > \frac{1}{2} \text{ and } y > \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases} \tag{1.4}$$

Every connected graph F satisfies $t(F,\overline{W}) = 2^{-v(F)}M(F,W)$, since a nonzero contribution requires all vertices of F to lie in the same half of $[0,1]$. Write $\sigma = M(P_2,W)$ and $\tau = M(K_3,W)$ as before. Applying Theorem 1.3 to $\overline{W}$ and using $M(K_2,W) = 1$, we obtain

$$\begin{aligned} 2^{-3k(n-1)}\tau^{k(n-1)} &= t(K_3,\overline{W})^{k(n-1)} \\ &\le t(T^{+k},\overline{W})\, t(K_2,\overline{W})^{n+k-3}\, t(P_2,\overline{W})^{(k-1)(n-2)} \\ &= 2^{-(n+k)-2(n+k-3)-3(k-1)(n-2)} M(T^{+k},W)\, \sigma^{(k-1)(n-2)} \\ &= 2^{-3k(n-1)} M(T^{+k},W)\, \sigma^{(k-1)(n-2)}. \end{aligned}$$

Dividing by $2^{-3k(n-1)}\sigma^{(k-1)(n-2)}$, which is positive since $\sigma \ge \frac{1}{2}$, gives (1.2). The details appear in Section 2.

Theorem 1.4 (Even-cycle apices)**.** *For every even $n \ge 4$, every integer $k \ge 1$, and every graphon W,*

$$M(C_n^{+k},W) \ge 2^{1-n(k+1)}. \tag{1.5}$$

If $k \ge 3$, then

$$M(C_n^{+k},W) \ge 2^{-nk} M(C_n,W) \ge 2^{1-n(k+1)}. \tag{1.6}$$

For $k \ge 3$, the relative bound (1.6) implies (1.5). Indeed, the even cycle C_n has the Sidorenko property and is therefore common, so $M(C_n,W) \ge 2^{1-n}$ and hence $2^{-nk}M(C_n,W) \ge 2^{1-n(k+1)}$. For $k = 1$ and $k = 2$, commonness of C_n^{+k} is known: C_n^{+1} is common by Sidorenko's theorem [15], and C_n^{+2} by [5, Theorem 4.3].

1.2 Proof outlines

The proof of Theorem 1.2 uses a conditional version of the branching random walks of Conlon, Kim, Lee, and Lee [1]. We bound $\hom(T^{+k},G)$ for a finite graph G from below by the entropy of a random homomorphism of T^{+k}, which we construct in three steps. We begin with a uniformly random triangle in G, whose entropy is bounded in terms of $\hom(K_2,G)$, $\hom(P_2,G)$, and $\hom(K_3,G)$ by Jensen's inequality. We then extend it to a random homomorphism of the book $K_{1,1,k}$, whose k pages are conditionally independent given the $K_{1,1}$ part; these pages will become the apices. We bound the entropies of the book from below using the chain rule (2.1) and subadditivity of mutual information (2.5). Finally, we extend the book to the whole tree by a random walk that reuses the law of the book on every edge of T; by the chain rule (2.1), the entropy of the result equals the entropy of the book plus $n-2$ times the entropy of the conditional distribution that this random walk gives.

For cycles, we use the positivity principle of Razborov's flag algebra method [12], which also underlies the computer-assisted part of [5]. We give it in a signed homomorphism-density basis: a nonnegative rooted quadratic form expands into a finite linear combination of signed graph densities. Three exact identities on six vertices provide the needed comparison at the fourth moment. The main analytic step converts this comparison into a statement that retains the larger of the two color moments. Convexity then raises the cycle length without losing the distinction between the colors.

Section 2 develops the entropy argument, and Section 3 proves the spectral and moment comparisons. The precise certificate data and verification are described in Appendix B. The results apply to two classes of connected Sidorenko graphs; they do not establish Conjecture 1.1 for arbitrary Sidorenko bases.

2 Tree apices through entropy

We first reduce Theorem 1.3 to a counting inequality for finite graphs, and then prove that inequality by the entropy method. For a finite simple graph G on N vertices, a *homomorphism*

from F to G is a map $V(F) \to V(G)$ sending every edge of F to an edge of G. Write $\mathrm{Hom}(F,G)$ for the set of homomorphisms, $\hom(F,G) \coloneqq |\mathrm{Hom}(F,G)|$, and the homomorphism density

$$t(F,G) \coloneqq \frac{\hom(F,G)}{N^{v(F)}}.$$

Every graphon W is a limit of finite simple graphs G_j, in the sense that $t(F,G_j) \to t(F,W)$ for every graph F; for instance, W-random graphs have this property almost surely [11]. Both sides of (1.3) are polynomials in four densities, so it suffices to prove (1.3) with W replaced by a finite simple graph G, which we fix for the rest of this section; let N be its number of vertices.

We bound $\hom(T^{+k},G)$ from below by constructing a random homomorphism from T^{+k} to G, whose entropy is a lower bound for $\log \hom(T^{+k},G)$. We construct it in three steps: we begin with a uniformly random triangle in G, then construct a random homomorphism of the book $K_{1,1,k}$ by adding k conditionally independent pages to an edge of the triangle, and finally extend the book along the tree.

2.1 Elementary facts on entropy

All random variables in this section take values in finite sets. For random variables A and B, write $p(a) \coloneqq \mathbb{P}(A=a)$, $p(a \mid b) \coloneqq \mathbb{P}(A=a \mid B=b)$, and $\mathrm{supp}(A) \coloneqq \{a : p(a) > 0\}$. The entropy of A and the conditional entropy of A given B are

$$\begin{aligned} \mathrm{H}(A) &\coloneqq -\sum_{a \in \mathrm{supp}(A)} p(a) \log p(a) = -\mathop{\mathbb{E}}_{A} \log p(A), \\ \mathrm{H}(A \mid B) &\coloneqq \sum_{b \in \mathrm{supp}(B)} p(b)\, \mathrm{H}(A \mid B=b) = -\mathop{\mathbb{E}}_{A,B} \log p(A \mid B), \end{aligned}$$

with the convention $0 \log 0 = 0$.

Lemma 2.1 (Elementary entropy identities)**.** *For random variables $A, B, C, B_1, \dots, B_k$,*

$$\begin{aligned} \mathrm{H}(A,B \mid C) &= \mathrm{H}(A \mid C) + \mathrm{H}(B \mid A,C), && \text{(chain rule, 2.1)} \\ \mathrm{H}(A) &\le \log |\mathrm{supp}(A)|, && \text{(support bound, 2.2)} \\ \mathrm{H}(A \mid B,C) &\le \mathrm{H}(A \mid C). && \text{(monotonicity under conditioning, 2.3)} \end{aligned}$$

The conditional mutual information, defined by

$$\mathrm{I}(A;B \mid C) \coloneqq \mathrm{H}(A \mid C) - \mathrm{H}(A \mid B,C) = \mathrm{H}(B \mid C) - \mathrm{H}(B \mid A,C), \tag{2.4}$$

is nonnegative. If $B_1, \dots, B_k$ are conditionally independent given (A,C), then

$$\mathrm{I}(A;B_1,\dots,B_k \mid C) \le \sum_{i=1}^{k} \mathrm{I}(A;B_i \mid C). \qquad \text{(subadditivity of mutual information, 2.5)}$$

Proof. On the support of (A,B,C) we have $p(a,b \mid c) = p(a \mid c)\, p(b \mid a,c)$, so

$$\begin{aligned} \mathrm{H}(A,B \mid C) &= -\mathop{\mathbb{E}}_{A,B,C} \log p(A,B \mid C) \\ &= -\mathop{\mathbb{E}}_{A,C} \log p(A \mid C) - \mathop{\mathbb{E}}_{A,B,C} \log p(B \mid A,C) \\ &= \mathrm{H}(A \mid C) + \mathrm{H}(B \mid A,C), \end{aligned}$$

which is (2.1).

Since log is concave, Jensen's inequality gives

$$\mathrm{H}(A) = \mathop{\mathbb{E}}_{A} \log \frac{1}{p(A)} \le \log \mathop{\mathbb{E}}_{A} \frac{1}{p(A)} = \log \sum_{a \in \operatorname{supp}(A)} p(a) \frac{1}{p(a)} = \log |\operatorname{supp}(A)|,$$

which is (2.2).

For (2.3), fix $c \in \operatorname{supp}(C)$, let $\phi(u) \coloneqq -u \log u$, which is concave on $[0,1]$, and let b range over the values with $p(b \mid c) > 0$. By the law of total probability,

$$p(a \mid c) = \sum_b p(b \mid c)\, p(a \mid b, c) \quad \text{for every } a, \qquad \sum_b p(b \mid c) = 1.$$

Applying Jensen's inequality to ϕ with the weights $p(b \mid c)$, for each a, gives

$$\begin{aligned} \mathrm{H}(A \mid C = c) &= \sum_a \phi\big(p(a \mid c)\big) = \sum_a \phi\Big(\sum_b p(b \mid c)\, p(a \mid b, c)\Big) \\ &\ge \sum_a \sum_b p(b \mid c)\, \phi\big(p(a \mid b, c)\big) = \sum_b p(b \mid c) \sum_a \phi\big(p(a \mid b, c)\big) \\ &= \sum_b p(b \mid c)\, \mathrm{H}(A \mid B = b, C = c). \end{aligned}$$

Multiplying by $p(c)$ and summing over $c \in \operatorname{supp}(C)$ gives

$$\mathrm{H}(A \mid C) = \sum_c p(c)\, \mathrm{H}(A \mid C = c) \ge \sum_{(b,c) \in \operatorname{supp}(B,C)} p(b,c)\, \mathrm{H}(A \mid B = b, C = c) = \mathrm{H}(A \mid B, C),$$

which is (2.3).

By the chain rule (2.1),

$$\mathrm{H}(A \mid C) + \mathrm{H}(B \mid A, C) = \mathrm{H}(A, B \mid C) = \mathrm{H}(B \mid C) + \mathrm{H}(A \mid B, C),$$

so the two expressions in (2.4) agree, and they are nonnegative by monotonicity under conditioning (2.3).

Finally, applying the chain rule (2.1) repeatedly, each time splitting off the first remaining variable, and then monotonicity under conditioning (2.3) gives

$$\begin{aligned} \mathrm{H}(B_1, \dots, B_k \mid C) &= \mathrm{H}(B_1 \mid C) + \mathrm{H}(B_2, \dots, B_k \mid B_1, C) \\ &= \mathrm{H}(B_1 \mid C) + \mathrm{H}(B_2 \mid B_1, C) + \mathrm{H}(B_3, \dots, B_k \mid B_1, B_2, C) \\ &= \cdots = \sum_{i=1}^{k} \mathrm{H}(B_i \mid B_1, \dots, B_{i-1}, C) \le \sum_{i=1}^{k} \mathrm{H}(B_i \mid C). \end{aligned}$$

Conditional independence means $p(b_1, \dots, b_k \mid a, c) = \prod_i p(b_i \mid a, c)$, so $\mathrm{H}(B_1, \dots, B_k \mid A, C) = \sum_i \mathrm{H}(B_i \mid A, C)$. Subtracting this from both sides of the previous inequality gives (2.5):

$$\begin{aligned} \mathrm{I}(A; B_1, \dots, B_k \mid C) &= \mathrm{H}(B_1, \dots, B_k \mid C) - \mathrm{H}(B_1, \dots, B_k \mid A, C) \\ &\le \sum_{i=1}^{k} \big(\mathrm{H}(B_i \mid C) - \mathrm{H}(B_i \mid A, C)\big) = \sum_{i=1}^{k} \mathrm{I}(A; B_i \mid C). \end{aligned}$$

□

2.2 A uniformly random triangle

For a vertex x, let $N_G(x)$ be its neighborhood and $d(x) \coloneqq |N_G(x)|$ its degree. For vertices x, y, let $r(x,y) \coloneqq |N_G(x) \cap N_G(y)|$ be their number of common neighbors. Write $\vec{E}(G)$ for the set of ordered pairs (x,y) with $xy \in E(G)$, and let

$$a(x) \coloneqq \bigl|\{(y,z) \in \vec{E}(G) : y, z \in N_G(x)\}\bigr|$$

be the number of ordered edges in the neighborhood of x. Then

$$\begin{aligned} &\hom(K_2, G) = |\vec{E}(G)|, \qquad \hom(P_2, G) = \sum_x d(x)^2, \\ &\hom(K_3, G) = \sum_{(x,y)\in\vec{E}(G)} r(x,y) = \sum_x a(x). \end{aligned} \tag{2.6}$$

Suppose that $\hom(K_3, G) > 0$, and let (X, Y, Z) be uniformly distributed on $\mathrm{Hom}(K_3, G)$, that is,

$$\mathbb{P}(X = x, Y = y, Z = z) = \frac{\mathbf{1}\{xy, xz, yz \in E(G)\}}{\hom(K_3, G)}.$$

This law is invariant under every permutation of X, Y, Z, and

$$\begin{aligned} &\mathrm{H}(X, Y, Z) = \log \hom(K_3, G), \\ &\mathbb{P}(X = x, Y = y) = \frac{r(x,y)}{\hom(K_3, G)} \quad ((x,y) \in \vec{E}(G)), \qquad \mathbb{P}(X = x) = \frac{a(x)}{\hom(K_3, G)}. \end{aligned} \tag{2.7}$$

Lemma 2.2 (Triangle entropy estimates)**.** *The uniformly random triangle (X, Y, Z) satisfies*

$$\mathrm{H}(Z \mid X, Y) \geq \log \frac{\hom(K_3, G)}{\hom(K_2, G)}, \qquad \mathrm{I}(Y; Z \mid X) \leq \log \frac{\hom(P_2, G)}{\hom(K_3, G)}, \tag{2.8}$$

and

$$\mathrm{H}(Y \mid X) - \mathrm{I}(Y; Z \mid X) = \mathrm{H}(Y \mid X, Z) = \mathrm{H}(Z \mid X, Y). \tag{2.9}$$

Proof. Given $(X, Y) = (x, y)$ with $r(x,y) > 0$, the vertex Z is uniform on $N_G(x) \cap N_G(y)$. Then

$$\begin{aligned} \mathrm{H}(Z \mid X, Y) &= \sum_{(x,y)\in\vec{E}(G)} \frac{r(x,y)}{\hom(K_3, G)} \log r(x,y) \\ &= \frac{\hom(K_2, G)}{\hom(K_3, G)} \cdot \frac{1}{\hom(K_2, G)} \sum_{(x,y)\in\vec{E}(G)} r(x,y) \log r(x,y) \\ &= \frac{\hom(K_2, G)}{\hom(K_3, G)} \mathop{\mathbb{E}}_{(X_0,Y_0)\sim\mathrm{Unif}[\vec{E}(G)]} \bigl[r(X_0, Y_0) \log r(X_0, Y_0)\bigr] \\ &\geq \frac{\hom(K_2, G)}{\hom(K_3, G)} \mathop{\mathbb{E}}_{(X_0,Y_0)\sim\mathrm{Unif}[\vec{E}(G)]} \bigl[r(X_0, Y_0)\bigr] \log \mathop{\mathbb{E}}_{(X_0,Y_0)\sim\mathrm{Unif}[\vec{E}(G)]} \bigl[r(X_0, Y_0)\bigr] \\ &= \frac{\hom(K_2, G)}{\hom(K_3, G)} \cdot \frac{\hom(K_3, G)}{\hom(K_2, G)} \log \frac{\hom(K_3, G)}{\hom(K_2, G)} = \log \frac{\hom(K_3, G)}{\hom(K_2, G)}. \end{aligned}$$

Here the first equality uses $\mathrm{H}(Z \mid X = x, Y = y) = \log r(x,y)$ and the distribution of (X, Y) in (2.7); the third uses $|\vec{E}(G)| = \hom(K_2, G)$; the inequality is Jensen's inequality for the convex function $u \log u$; and the last line uses $\mathbb{E}_{(X_0,Y_0)\sim\mathrm{Unif}[\vec{E}(G)]}[r(X_0, Y_0)] = \hom(K_3, G)/\hom(K_2, G)$, which follows from (2.6).

For the second bound, if $a(x) > 0$, then (Y, Z) given $X = x$ is uniform on a set of size $a(x)$, and both coordinates lie in $N_G(x)$. Hence

$$\begin{aligned}\mathrm{I}(Y; Z \mid X = x) &= \mathrm{H}(Y \mid X = x) - \mathrm{H}(Y \mid Z, X = x) \\ &= \mathrm{H}(Y \mid X = x) - \big(\mathrm{H}(Y, Z \mid X = x) - \mathrm{H}(Z \mid X = x)\big) \\ &= \mathrm{H}(Y \mid X = x) + \mathrm{H}(Z \mid X = x) - \mathrm{H}(Y, Z \mid X = x) \\ &\le 2 \log d(x) - \log a(x),\end{aligned}$$

where the first equality is the definition of mutual information (2.4) conditionally on $X = x$, the second is the chain rule (2.1) conditionally on $X = x$, and the inequality uses $\mathrm{H}(Y \mid X = x), \mathrm{H}(Z \mid X = x) \le \log d(x)$ by the support bound (2.2) and $\mathrm{H}(Y, Z \mid X = x) = \log a(x)$ by uniformity. Averaging over X gives

$$\begin{aligned}\mathrm{I}(Y; Z \mid X) &= \sum_{x \in \mathrm{supp}(X)} \mathbb{P}(X = x)\, \mathrm{I}(Y; Z \mid X = x) = \sum_{x : a(x) > 0} \frac{a(x)}{\hom(K_3, G)}\, \mathrm{I}(Y; Z \mid X = x) \\ &\le \sum_{x : a(x) > 0} \frac{a(x)}{\hom(K_3, G)} \log \frac{d(x)^2}{a(x)} \le \log\Big(\sum_{x : a(x) > 0} \frac{a(x)}{\hom(K_3, G)} \cdot \frac{d(x)^2}{a(x)} \Big) \\ &= \log \sum_{x : a(x) > 0} \frac{d(x)^2}{\hom(K_3, G)} \le \log \sum_{x} \frac{d(x)^2}{\hom(K_3, G)} = \log \frac{\hom(P_2, G)}{\hom(K_3, G)},\end{aligned}$$

where the first equality follows from the definitions of conditional entropy and mutual information (2.4); the second uses $\mathbb{P}(X = x) = a(x)/\hom(K_3, G)$ from (2.7); the first inequality is the bound $\mathrm{I}(Y; Z \mid X = x) \le 2 \log d(x) - \log a(x)$ above; the second inequality is Jensen's inequality for the concave function log, with the weights $a(x)/\hom(K_3, G)$, which sum to one by (2.6); and the last equality is $\sum_x d(x)^2 = \hom(P_2, G)$ from (2.6). This proves (2.8).

For (2.9), the first equality is the definition of mutual information (2.4), and the second holds because (X, Y, Z) and (X, Z, Y) have the same distribution, the law of the uniformly random triangle being invariant under permutations of the coordinates. □

2.3 Conditionally independent pages of a book

For an integer $k \ge 1$, let Q_k be the law of $(X, Y, Z_1, \dots, Z_k)$ obtained by sampling (X, Y) as in the uniformly random triangle (X, Y, Z), and then $Z_1, \dots, Z_k$ conditionally independently given (X, Y), each with the same conditional distribution as Z, that is,

$$Q_k(x, y, z_1, \dots, z_k) \coloneqq \underbrace{\frac{r(x, y)}{\hom(K_3, G)}}_{\text{Law of } (X,Y)} \prod_{i=1}^{k} \underbrace{\frac{\mathbf{1}\{z_i \in N_G(x) \cap N_G(y)\}}{r(x, y)}}_{\text{Law of } (Z_i \mid X,Y)} \qquad ((x, y) \in \vec{E}(G)),$$

and $Q_k = 0$ otherwise. Then Q_k is supported on homomorphisms of the book $K_{1,1,k}$, every triple (X, Y, Z_i) has the law of the uniformly random triangle, and Q_k is invariant under exchanging X and Y.

Lemma 2.3 (Entropy of the book law)**.** *Let $Z_{1:k} \coloneqq (Z_1, \dots, Z_k)$. Then*

$$\mathrm{H}_{Q_k}(Z_{1:k}, X, Y) \ge k \log \hom(K_3, G) - (k - 1) \log \hom(K_2, G), \tag{2.10}$$

$$\mathrm{H}_{Q_k}(Y \mid Z_{1:k}, X) \ge k \log \hom(K_3, G) - \log \hom(K_2, G) - (k - 1) \log \hom(P_2, G). \tag{2.11}$$

Proof. For the first bound, we first separate the pages:

$$\begin{aligned}\mathrm{H}_{Q_k}(Z_{1:k}, X, Y) &= \mathrm{H}(X, Y) + \mathrm{H}_{Q_k}(Z_{1:k} \mid X, Y) = \mathrm{H}(X, Y) + \sum_{i=1}^{k} \mathrm{H}_{Q_k}(Z_i \mid X, Y) \\ &= \mathrm{H}(X, Y) + k\, \mathrm{H}(Z \mid X, Y),\end{aligned}$$

where the first equality is the chain rule (2.1), the second holds because $Z_1, \dots, Z_k$ are conditionally independent given (X, Y), and the third because every (X, Y, Z_i) has the law of the uniformly random triangle (X, Y, Z). We then bound the result:

$$\begin{aligned}\mathrm{H}(X,Y) + k\,\mathrm{H}(Z \mid X,Y) &= \mathrm{H}(X,Y,Z) + (k-1)\mathrm{H}(Z \mid X,Y) \\ &= \log \hom(K_3, G) + (k-1)\mathrm{H}(Z \mid X,Y) \\ &\geq \log \hom(K_3, G) + (k-1)\log \frac{\hom(K_3,G)}{\hom(K_2,G)},\end{aligned}$$

where the first equality is the chain rule (2.1) for (X, Y, Z), the second is (2.7), and the inequality is (2.8). This proves (2.10).

For the second bound, we first separate the pages:

$$\begin{aligned}\mathrm{H}_{Q_k}(Y \mid Z_{1:k}, X) &= \mathrm{H}_{Q_k}(Y \mid X) - \mathrm{I}_{Q_k}(Y; Z_{1:k} \mid X) \geq \mathrm{H}_{Q_k}(Y \mid X) - \sum_{i=1}^{k} \mathrm{I}_{Q_k}(Y; Z_i \mid X) \\ &= \mathrm{H}(Y \mid X) - k\,\mathrm{I}(Y; Z \mid X),\end{aligned}$$

where the first equality is the definition of mutual information (2.4); the inequality is the subadditivity of mutual information (2.5) with $A = Y$, $C = X$, and $B_i = Z_i$, which holds because $Z_1, \dots, Z_k$ are conditionally independent given (X, Y); and the second equality holds because every (X, Y, Z_i) has the law of the uniformly random triangle (X, Y, Z). We then bound the result:

$$\begin{aligned}\mathrm{H}(Y \mid X) - k\,\mathrm{I}(Y; Z \mid X) &= \mathrm{H}(Z \mid X,Y) - (k-1)\,\mathrm{I}(Y; Z \mid X) \\ &\geq \log \frac{\hom(K_3,G)}{\hom(K_2,G)} - (k-1)\log \frac{\hom(P_2,G)}{\hom(K_3,G)},\end{aligned}$$

where the equality is (2.9) and the inequality is (2.8). This proves (2.11). □

2.4 Extension along the tree

Finally, we extend the book along the tree. We first show this for general random variables (S, X, Y) whose distribution is unchanged by exchanging X and Y. The following lemma is a conditional version of the random walk used in [1].

Lemma 2.4 (Extension along a tree)**.** *Let T be a tree on $n \geq 2$ vertices. For every joint law Q of random variables (S, X, Y), where X and Y take values in the same set and*

$$Q(S = s, X = x, Y = y) = Q(S = s, X = y, Y = x) \quad \text{for all } s, x, y, \tag{2.12}$$

there is a joint law P_Q of (S, X_T), where $X_T := (X_v)_{v \in V(T)}$, such that, for every edge uv of T, the law of (S, X_u, X_v) under P_Q is Q, and

$$\mathrm{H}_{P_Q}(S, X_T) = \mathrm{H}_Q(S, X, Y) + (n-2)\mathrm{H}_Q(Y \mid S, X). \tag{2.13}$$

Proof. For each s with $Q(S = s) > 0$, let $\pi_s(x, y) := Q(X = x, Y = y \mid S = s)$ and $\mu_s(x) := \sum_y \pi_s(x, y)$. By (2.12), π_s is symmetric, so both of its marginals are μ_s. Define the transition matrix

$$K_s(x, y) := \begin{cases} \pi_s(x,y)/\mu_s(x), & \mu_s(x) > 0, \\ \mu_s(y), & \mu_s(x) = 0. \end{cases}$$

Every row sums to one, and μ_s is a stationary distribution of K_s:

$$\sum_x \mu_s(x) K_s(x, y) = \sum_x \pi_s(x, y) = \mu_s(y). \tag{2.14}$$

Fix a vertex r of T. Every vertex $v \neq r$ has a unique neighbor on the path from v to r, called its parent, and we write $u \to v$ if u is the parent of v. Every edge of T is written as $u \to v$ for exactly one order of its endpoints. Define

$$P_Q(S = s, X_T = x_T) \coloneqq Q(S = s)\, \mu_s(x_r) \prod_{u \to v} K_s(x_u, x_v), \tag{2.15}$$

where $x_T = (x_v)_{v \in V(T)}$. To show that P_Q is a probability distribution, let $v \neq r$ be a leaf of T with parent u. Since $\sum_y K_s(x_u, y) = 1$, summing (2.15) over x_v gives the same formula for the tree $T - v$. Repeating this until only r remains leaves $\sum_s Q(S = s) \sum_x \mu_s(x) = 1$. Note that the same computation shows that for every subtree T' containing r, the law of $(S, X_{T'})$ under P_Q is given by (2.15) with T' in place of T.

By (2.14) and induction on the distance from the root, every X_v has law μ_s given $S = s$. Hence for every edge $u \to v$,

$$P_Q(X_u = x, X_v = y \mid S = s) = \mu_s(x) K_s(x, y) = \pi_s(x, y),$$

which also holds when $\mu_s(x) = 0$, since both sides then vanish. Averaging over s shows that for every edge $u \to v$, the triple (S, X_u, X_v) has law Q, whereas by the symmetry (2.12) so does (S, X_v, X_u).

Order all vertices of T as $r = v_1, v_2, \ldots, v_n$ so that every parent comes before its children. For $j \geq 2$, the parent u of v_j is then v_i for some $i < j$. As $v_1, \ldots, v_j$ span a subtree containing r, the note above gives

$$P_Q\big(X_{v_j} = y \mid S = s,\ X_{v_i} = x_{v_i} \text{ for } i < j\big) = K_s(x_u, y),$$

so $\mathrm{H}_{P_Q}(X_{v_j} \mid S, X_{v_1}, \ldots, X_{v_{j-1}}) = \mathrm{H}_{P_Q}(X_{v_j} \mid S, X_u) = \mathrm{H}_Q(Y \mid S, X)$, where the last equality holds because (S, X_u, X_{v_j}) has law Q. Therefore

$$\begin{aligned} \mathrm{H}_{P_Q}(S, X_T) &= \mathrm{H}_{P_Q}(S) + \mathrm{H}_{P_Q}(X_{v_1} \mid S) + \sum_{j=2}^{n} \mathrm{H}_{P_Q}(X_{v_j} \mid S, X_{v_1}, \ldots, X_{v_{j-1}}) \\ &= \mathrm{H}_Q(S) + \mathrm{H}_Q(X \mid S) + (n-1)\mathrm{H}_Q(Y \mid S, X) \\ &= \mathrm{H}_Q(S, X, Y) + (n-2)\mathrm{H}_Q(Y \mid S, X), \end{aligned}$$

where the first equality is the chain rule (2.1), applied repeatedly; the second holds because $(S, X_{v_1}) = (S, X_r)$ has the same law under P_Q as (S, X) under Q, together with the identity above for each $j \geq 2$; and the third is the chain rule (2.1) for (S, X, Y), that is, $\mathrm{H}_Q(S, X, Y) = \mathrm{H}_Q(S) + \mathrm{H}_Q(X \mid S) + \mathrm{H}_Q(Y \mid S, X)$. □

2.5 The counting inequality

Proposition 2.5. *For every tree T on $n \geq 2$ vertices and every $k \geq 1$,*

$$\hom(T^{+k}, G)\, \hom(K_2, G)^{n+k-3}\, \hom(P_2, G)^{(k-1)(n-2)} \geq \hom(K_3, G)^{k(n-1)}. \tag{2.16}$$

Proof. If $\hom(K_3, G) = 0$, the right side vanishes. So we can assume that $\hom(K_3, G) > 0$, which implies that $\hom(K_2, G), \hom(P_2, G) > 0$ as well. Since Q_k is invariant under exchanging X and Y, we may let P_{Q_k} be the law of $(Z_{1:k}, X_T)$ constructed by Lemma 2.4 with $S = Z_{1:k}$, and its marginal on $(Z_{1:k}, X_u, X_v)$ is Q_k for every edge uv of T.

Let $(z_{1:k}, x_T)$ be in the support of P_{Q_k}. For every edge uv of T, the tuple $(z_{1:k}, x_u, x_v)$ is in the support of Q_k, so $x_u x_v \in E(G)$ and every z_i is adjacent to both x_u and x_v, so every

x_v is adjacent to every z_i. Hence $(z_{1:k}, x_T)$ is a homomorphism of T^{+k} into G, that is, P_{Q_k} is supported on $\mathrm{Hom}(T^{+k}, G)$. Therefore

$$\begin{aligned}
\log \hom(T^{+k}, G) &\geq \mathrm{H}_{P_{Q_k}}(Z_{1:k}, X_T) = \mathrm{H}_{Q_k}(Z_{1:k}, X, Y) + (n-2)\mathrm{H}_{Q_k}(Y \mid Z_{1:k}, X) \\
&\geq \big(k \log \hom(K_3, G) - (k-1) \log \hom(K_2, G)\big) \\
&\qquad + (n-2)\big(k \log \hom(K_3, G) - \log \hom(K_2, G) - (k-1) \log \hom(P_2, G)\big) \\
&= k(n-1) \log \hom(K_3, G) - (n+k-3) \log \hom(K_2, G) \\
&\qquad - (k-1)(n-2) \log \hom(P_2, G),
\end{aligned}$$

where the first inequality is the support bound (2.2), the equality is (2.13), the second inequality is Lemma 2.3 together with $n - 2 \geq 0$, and the last equality collects terms. Exponentiating proves (2.16). □

Proof of Theorem 1.3. By the reduction at the beginning of this section, we may replace W by G. Then

$$\begin{aligned}
&t(T^{+k}, G)\, t(K_2, G)^{n+k-3}\, t(P_2, G)^{(k-1)(n-2)} \\
&\quad = \frac{\hom(T^{+k}, G)\ \hom(K_2, G)^{n+k-3}\ \hom(P_2, G)^{(k-1)(n-2)}}{N^{(n+k)+2(n+k-3)+3(k-1)(n-2)}} \\
&\quad = \frac{\hom(T^{+k}, G)\ \hom(K_2, G)^{n+k-3}\ \hom(P_2, G)^{(k-1)(n-2)}}{N^{3k(n-1)}} \\
&\quad \geq \frac{\hom(K_3, G)^{k(n-1)}}{N^{3k(n-1)}} = t(K_3, G)^{k(n-1)},
\end{aligned}$$

where the first equality holds because T^{+k}, K_2, P_2, and K_3 have $n + k$, 2, 3, and 3 vertices, respectively; the second is $(n + k) + 2(n + k - 3) + 3(k - 1)(n - 2) = 3k(n - 1)$; and the inequality is (2.16). This proves (1.3). □

3 Even-cycle apices through fourth moments

For $k = 1$ and $k = 2$, the commonness of C_n^{+k} is known [15], [5, Theorem 4.3]. For $k \geq 3$, Hölder's inequality reduces the relative bound (1.6) to three apices.

Lemma 3.1 (Apex lifting)**.** *For every graphon W, every $n \geq 3$, and all $k \geq s \geq 1$,*

$$M(C_n^{+k}, W)\, M(C_n, W)^{k/s-1} \geq M(C_n^{+s}, W)^{k/s}. \tag{3.1}$$

Proof. Let $L \in \{W, 1 - W\}$, and write $\mathbf{x} = (x_0, \ldots, x_{n-1}) \in [0,1]^n$ with $x_n = x_0$. Integrating out the j apices one at a time gives

$$t(C_n^{+j}, L) = \int_{[0,1]^n} \prod_{i=0}^{n-1} L(x_i, x_{i+1}) \left(\int_0^1 \prod_{i=0}^{n-1} L(z, x_i)\, \mathrm{d}z \right)^j \mathrm{d}\mathbf{x} \qquad (j \geq 0).$$

Therefore

$$\begin{aligned}
&M(C_n^{+s}, W) \\
&= \sum_L \int \left(\prod_{i=0}^{n-1} L(x_i, x_{i+1}) \left(\int \prod_{i=0}^{n-1} L(z, x_i)\, \mathrm{d}z \right)^k \right)^{s/k} \left(\prod_{i=0}^{n-1} L(x_i, x_{i+1}) \right)^{1-s/k} \mathrm{d}\mathbf{x} \\
&\leq \left(\sum_L \int \prod_{i=0}^{n-1} L(x_i, x_{i+1}) \left(\int \prod_{i=0}^{n-1} L(z, x_i)\, \mathrm{d}z \right)^k \mathrm{d}\mathbf{x} \right)^{s/k} \left(\sum_L \int \prod_{i=0}^{n-1} L(x_i, x_{i+1})\, \mathrm{d}\mathbf{x} \right)^{1-s/k} \\
&= M(C_n^{+k}, W)^{s/k} M(C_n, W)^{1-s/k},
\end{aligned}$$

where the sums are over $L \in \{W, 1-W\}$, and the inequality is Hölder's inequality with exponents k/s and $k/(k-s)$ for the sum over L and the integral over $\mathbf{x}$. Raising both sides to the power k/s gives (3.1). □

Suppose that $n \geq 4$ is even and that W satisfies $M(C_n^{+3}, W) \geq 2^{-3n} M(C_n, W)$. For $k \geq 3$, Lemma 3.1 with $s = 3$ gives

$$M(C_n^{+k}, W)\, M(C_n, W)^{k/3-1} \geq M(C_n^{+3}, W)^{k/3} \geq 2^{-nk} M(C_n, W)^{k/3},$$

and dividing by $M(C_n, W)^{k/3-1}$, which is positive since C_n is common, gives

$$M(C_n^{+k}, W) \geq 2^{-nk} M(C_n, W) \qquad (k \geq 3). \tag{3.2}$$

Therefore it suffices to prove the three-apex inequality

$$M(C_n^{+3}, W) \geq 2^{-3n} M(C_n, W).$$

A graphon W is a *step graphon* if, for some $d \geq 1$, it is constant on each $I_\alpha \times I_\beta$, where $I_\alpha \coloneqq \left(\frac{\alpha-1}{d}, \frac{\alpha}{d}\right]$ for $\alpha \in [d]$.

Lemma 3.2 (Reduction to step graphons)**.** *If* (3.2) *holds for every step graphon, then it holds for every graphon.*

The proof is given in Appendix A. We will also often use the following inequality on random variables.

Lemma 3.3 (Moment inequality)**.** *For a nonnegative bounded random variable X and $1 \leq p \leq q$,*

$$\mathop{\mathbb{E}}_X X^q \geq \left(\mathop{\mathbb{E}}_X X^p\right)^{q/p}. \tag{3.3}$$

Proof. Since $q/p \geq 1$, the function $y \mapsto y^{q/p}$ is convex on $[0, \infty)$, and Jensen's inequality applied to X^p gives

$$\mathop{\mathbb{E}}_X X^q = \mathop{\mathbb{E}}_X (X^p)^{q/p} \geq \left(\mathop{\mathbb{E}}_X X^p\right)^{q/p}. \qquad \square$$

We prove the three-apex inequality in three steps. In Section 3.1 we fix a color and the positions of the three apices, and bound the density of C_n^{+3} from below by a power of the mean square degree inside the common neighborhood of the apices (Lemma 3.5). In Section 3.2 we bound the expected square of this mean square degree from below by the average $8M(C_4, W)$ of the four-cycle densities of the two colors (Lemma 3.7), and in Section 3.3 we bound a weighted expected square from below by the larger of them (Proposition 3.12), and then combine these bounds.

3.1 Conditioning on the apices

By Lemma 3.2, it suffices to prove the three-apex inequality when W is a step graphon. From now on, we fix $d \geq 1$ and assume that W is constant on each $I_\alpha \times I_\beta$, where $I_\alpha = \left(\frac{\alpha-1}{d}, \frac{\alpha}{d}\right]$ for $\alpha \in [d]$.

A *kernel* is a bounded symmetric measurable function $L : [0,1]^2 \to \mathbb{R}$, and $t(F, L)$ is defined by the same formula as for graphons. For a kernel L that is constant on each $I_\alpha \times I_\beta$, like W, let $T_L \in \mathbb{R}^{d \times d}$ be the matrix with $(T_L)_{\alpha\beta} \coloneqq \frac{1}{d} L(x, y)$ for $(x, y) \in I_\alpha \times I_\beta$. We use the inner product $\langle x, y \rangle \coloneqq d^{-1} \sum_\alpha x_\alpha y_\alpha$ on $\mathbb{R}^d$, so that the constant vector $\mathbf{1}$ has norm one.

Lemma 3.4 (Traces)**.** *Let L be a kernel that is constant on each $I_\alpha \times I_\beta$. For $n \geq 3$,*

$$t(C_n, L) = \operatorname{Tr} T_L^n. \tag{3.4}$$

If $n \geq 4$ is even and $g \in \mathbb{R}^d$ is a unit vector, then

$$t(C_n, L) \geq \|T_L g\|^n \geq |\langle g, T_L g\rangle|^n, \qquad t(C_n, L) \leq t(C_4, L)^{n/4}. \tag{3.5}$$

Proof. Label the vertices of C_n by $1, \dots, n$ in cyclic order, and write $x_{n+1} \coloneqq x_1$ and $\alpha_{n+1} \coloneqq \alpha_1$. Then

$$\begin{aligned} t(C_n, L) &= \int_{[0,1]^n} \prod_{j=1}^{n} L(x_j, x_{j+1}) \, \mathrm{d}\mathbf{x} = \sum_{\alpha_1, \dots, \alpha_n \in [d]} \int_{I_{\alpha_1} \times \cdots \times I_{\alpha_n}} \prod_{j=1}^{n} L(x_j, x_{j+1}) \, \mathrm{d}\mathbf{x} \\ &= \sum_{\alpha_1, \dots, \alpha_n \in [d]} \prod_{j=1}^{n} (T_L)_{\alpha_j \alpha_{j+1}} = \operatorname{Tr} T_L^n, \end{aligned}$$

where the second equality splits $[0,1]^n$ into the boxes $I_{\alpha_1} \times \cdots \times I_{\alpha_n}$; the third holds because on such a box each factor $L(x_j, x_{j+1})$ equals $d\,(T_L)_{\alpha_j \alpha_{j+1}}$ and the box has volume d^{-n}; and the last is the expansion of the trace of a matrix power.

Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of the symmetric matrix T_L with an orthonormal basis of eigenvectors $e_1, \dots, e_d$, and write $g = \sum_i c_i e_i$ with $\sum_i c_i^2 = 1$. Since n is even,

$$\|T_L g\|^n = \big(\|T_L g\|^2\big)^{n/2} = \Big(\sum_i \lambda_i^2 c_i^2\Big)^{n/2} \leq \max_i |\lambda_i|^n \leq \sum_i \lambda_i^n = \operatorname{Tr} T_L^n = t(C_n, L),$$

and $|\langle g, T_L g\rangle| \leq \|T_L g\|$ by Cauchy–Schwarz. This proves the first part of (3.5). Finally,

$$\begin{aligned} t(C_n, L) &= \sum_i \lambda_i^4 \, \lambda_i^{n-4} \leq \max_i \lambda_i^{n-4} \sum_i \lambda_i^4 \\ &\leq \Big(\sum_i \lambda_i^4\Big)^{(n-4)/4} \sum_i \lambda_i^4 = t(C_4, L)^{n/4}, \end{aligned}$$

where the first inequality holds since $\lambda_i^4 \geq 0$, and the second since $\max_i \lambda_i^{n-4} = (\max_i \lambda_i^4)^{(n-4)/4}$ and $\max_i \lambda_i^4 \leq \sum_i \lambda_i^4$; both use that n is even. □

We write the Ramsey multiplicities as expectations over a uniformly random sign $\varepsilon \in \{\pm 1\}$. Let $U \coloneqq 2W - 1$ and $S_\varepsilon \coloneqq 1 + \varepsilon U$ be kernels, so that $S_{+1} = 2W$ and $S_{-1} = 2(1 - W)$. Note that these kernels are constant on each $I_\alpha \times I_\beta$. Then

$$\mathop{\mathbb{E}}_{\varepsilon} t(F, S_\varepsilon) = 2^{e(F)-1} M(F, W). \tag{3.6}$$

For a graph F and $A \subseteq E(F)$, let F_A be the graph with vertex set $V(F)$ and edge set A. Then

$$\begin{aligned} t(F, S_\varepsilon) &= \int_{[0,1]^{V(F)}} \prod_{uv \in E(F)} \big(1 + \varepsilon U(x_u, x_v)\big) \, \mathrm{d}\mathbf{x} \\ &= \sum_{A \subseteq E(F)} \varepsilon^{|A|} \int_{[0,1]^{V(F)}} \prod_{uv \in A} U(x_u, x_v) \, \mathrm{d}\mathbf{x} = \sum_{A \subseteq E(F)} \varepsilon^{|A|} t(F_A, U), \end{aligned} \tag{3.7}$$

where the second equality chooses, for each edge uv, either 1 or $\varepsilon U(x_u, x_v)$. Since $\mathbb{E}_\varepsilon \, \varepsilon^{|A|}$ is 1 for even $|A|$ and 0 for odd $|A|$,

$$\mathop{\mathbb{E}}_{\varepsilon} t(F, S_\varepsilon) = \sum_{\substack{A \subseteq E(F) \\ |A| \text{ even}}} t(F_A, U).$$

We now condition on the positions of the apices: for $\mathbf{z} = (z_1, z_2, z_3) \in [0,1]^3$, let

$$t_{\mathbf{z}}(F^{+3}, L) \coloneqq \int_{[0,1]^{V(F)}} \prod_{uv \in E(F)} L(x_u, x_v) \prod_{v \in V(F)} \prod_{i=1}^{3} L(z_i, x_v)\, \mathrm{d}\mathbf{x}$$

be the density of F^{+3} with the apices placed at z_1, z_2, z_3. Averaging this conditional density over $\mathbf{z} \sim \mathrm{Unif}[0,1]^3$ gives $t(F^{+3}, S_\varepsilon)$, so (3.6) gives

$$\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}} t_{\mathbf{z}}(F^{+3}, S_\varepsilon) = 2^{e(F^{+3})-1} M(F^{+3}, W). \tag{3.8}$$

With this conditioning, we can view $t_{\mathbf{z}}(C_n^{+3}, S_\varepsilon)$ as the density $t(C_n, L)$ of a kernel L and apply Lemma 3.4 to L, which gives the following lower bound.

Lemma 3.5 (Conditioning)**.** *For every $\varepsilon \in \{\pm 1\}$, every $\mathbf{z} \in [0,1]^3$, and every even $n \geq 4$,*

$$t_{\mathbf{z}}(C_n^{+3}, S_\varepsilon) \geq \left(\frac{t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)} \right)^{n/2}. \tag{3.9}$$

Here the ratio is interpreted as 0 when $t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon) = 0$.

Proof. Let $L_{\varepsilon,\mathbf{z}}$ be the kernel S_ε weighted by the common neighborhood of the apices,

$$L_{\varepsilon,\mathbf{z}}(x,y) \coloneqq \Big(\prod_{i=1}^{3} S_\varepsilon(z_i, x) S_\varepsilon(z_i, y) \Big)^{1/2} S_\varepsilon(x,y),$$

which is constant on each $I_\alpha \times I_\beta$. With $x_{n+1} \coloneqq x_1$,

$$\begin{aligned} t(C_n, L_{\varepsilon,\mathbf{z}}) &= \int_{[0,1]^n} \prod_{j=1}^{n} S_\varepsilon(x_j, x_{j+1}) \prod_{i=1}^{3} \prod_{j=1}^{n} \big(S_\varepsilon(z_i, x_j) S_\varepsilon(z_i, x_{j+1}) \big)^{1/2} \mathrm{d}\mathbf{x} \\ &= \int_{[0,1]^n} \prod_{j=1}^{n} S_\varepsilon(x_j, x_{j+1}) \prod_{i=1}^{3} \prod_{j=1}^{n} S_\varepsilon(z_i, x_j)\, \mathrm{d}\mathbf{x} = t_{\mathbf{z}}(C_n^{+3}, S_\varepsilon), \end{aligned}$$

where the second equality holds since $\prod_{j=1}^{n} S_\varepsilon(z_i, x_{j+1}) = \prod_{j=1}^{n} S_\varepsilon(z_i, x_j)$ for each i. Suppose that $t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon) > 0$, and let $g_{\varepsilon,\mathbf{z}} \in \mathbb{R}^d$ be the vector with

$$(g_{\varepsilon,\mathbf{z}})_\alpha \coloneqq \left(\frac{\prod_{i=1}^{3} S_\varepsilon(z_i, x)}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)} \right)^{1/2} \qquad (x \in I_\alpha).$$

Then $g_{\varepsilon,\mathbf{z}}$ is a unit vector, since

$$\|g_{\varepsilon,\mathbf{z}}\|^2 = \frac{1}{d} \sum_{\alpha \in [d]} (g_{\varepsilon,\mathbf{z}})_\alpha^2 = \frac{1}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)} \int_0^1 \prod_{i=1}^{3} S_\varepsilon(z_i, x)\, \mathrm{d}x = 1,$$

where the second equality holds since $\prod_{i=1}^{3} S_\varepsilon(z_i, x)$ is constant in $x \in I_\alpha$ and I_α has length $1/d$, and the last is the definition of $t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)$. In the same way,

$$\|T_{L_{\varepsilon,\mathbf{z}}} g_{\varepsilon,\mathbf{z}}\|^2 = \frac{1}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)} \int_0^1 \prod_{i=1}^{3} S_\varepsilon(z_i, x) \Big(\int_0^1 S_\varepsilon(x,y) \prod_{i=1}^{3} S_\varepsilon(z_i, y)\, \mathrm{d}y \Big)^2 \mathrm{d}x = \frac{t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)}.$$

Therefore

$$t_{\mathbf{z}}(C_n^{+3}, S_\varepsilon) = t(C_n, L_{\varepsilon,\mathbf{z}}) \geq \|T_{L_{\varepsilon,\mathbf{z}}} g_{\varepsilon,\mathbf{z}}\|^n = \Big(\|T_{L_{\varepsilon,\mathbf{z}}} g_{\varepsilon,\mathbf{z}}\|^2 \Big)^{n/2} = \left(\frac{t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)} \right)^{n/2},$$

where the inequality is (3.5) for the kernel $L_{\varepsilon,\mathbf{z}}$ and the unit vector $g_{\varepsilon,\mathbf{z}}$. This proves (3.9) when $t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon) > 0$.

If $t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon) = 0$, then $\int_0^1 \prod_{i=1}^{3} S_\varepsilon(z_i, x)\, \mathrm{d}x = 0$. Since $S_\varepsilon \geq 0$, this means that $\prod_{i=1}^{3} S_\varepsilon(z_i, x) = 0$ for almost every x. Hence $L_{\varepsilon,\mathbf{z}}(x,y) = 0$ for almost every (x,y), so $t_{\mathbf{z}}(C_n^{+3}, S_\varepsilon) = t(C_n, L_{\varepsilon,\mathbf{z}}) = 0$, and both sides of (3.9) vanish. □

3.2 The fourth moment

In this subsection, we bound

$$\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}\left(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}\right)^2$$

from below by $8M(C_4,W)$.

For the proof, we use the following lemma, which is proven with a computational certificate. Together with Lemma 3.11, it is the only computer-assisted input of the proof.

Lemma 3.6 (Certificate for the mean)**.** *For every graphon W,*

$$M(K_{1,2,3},W)\geq\frac{M(K_{2,3},W)}{32}. \tag{3.10}$$

Proof. This inequality is proven by a flag algebra certificate, where the specification of this problem is described in Appendix B.1. □

Lemma 3.7 (Unweighted comparison)**.** *For any graphon W, recall that $S_\varepsilon=1+\varepsilon U$, where $U=2W-1$. Then*

$$\begin{aligned}\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}\left(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}\right)^2&\geq 2^{11}M(K_{1,2,3},W)-2^5M(K_{2,3},W)\\&\geq\big(8M(C_4,W)\big)^{3/2}\geq 8M(C_4,W).\end{aligned} \tag{3.11}$$

Proof. To handle the ratio, we simplify it with the following bound: for every $\varepsilon\in\{\pm1\}$ and every $\mathbf{z}\in[0,1]^3$,

$$\left(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}\right)^2\geq 2t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)-t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2. \tag{3.12}$$

To see why, if $t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)>0$, then

$$\left(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}\right)^2-2t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)+t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2=\left(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}-t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)\right)^2\geq 0,$$

and if $t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)=0$, then both sides of (3.12) vanish. Hence the left side of (3.11) is at least

$$\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}\big(2t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)-t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2\big).$$

For a kernel L, expanding the square as an integral over two independent vertices gives

$$t_{\mathbf{z}}(K_1^{+3},L)^2=\Big(\int_0^1\prod_{i=1}^3L(z_i,x)\,\mathrm{d}x\Big)^2=\int_{[0,1]^2}\prod_{i=1}^3L(z_i,x)L(z_i,y)\,\mathrm{d}x\,\mathrm{d}y=t_{\mathbf{z}}(\overline{K_2}^{+3},L),$$

where $\overline{K_2}$ is the graph with two vertices and no edges. Applying (3.8) with $F=\overline{K_2}$ and $F=P_2$ gives

$$\begin{aligned}\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2&=\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}t_{\mathbf{z}}(\overline{K_2}^{+3},S_\varepsilon)=2^5M(\overline{K_2}^{+3},W),\\\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)&=2^{10}M(P_2^{+3},W),\end{aligned}$$

where we used $e(\overline{K_2}^{+3})=6$ and $e(P_2^{+3})=11$. Since $\overline{K_2}^{+3}=K_{2,3}$ and $P_2^{+3}=K_{1,2,3}$, we have

$$\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2=2^5M(K_{2,3},W),\qquad\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)=2^{10}M(K_{1,2,3},W), \tag{3.13}$$

which proves the first inequality of (3.11). By (3.10),

$$2^{11}M(K_{1,2,3},W) - 2^5 M(K_{2,3},W) \ge 2^5 M(K_{2,3},W) = \mathop{\mathbb{E}}_{\varepsilon} t(K_{2,3},S_\varepsilon),$$

where the equality is (3.6).

Next, we view $\mathbb{E}_\varepsilon\, t(K_{2,3},S_\varepsilon)$ as the third moment of a random variable of ε and the part with two vertices. Then the second moment of the same random variable is $8M(C_4,W)$, which allows us to use the moment inequality (3.3) to prove the inequality. Specifically, let X,Y,Z be independent and uniform on $[0,1]$. Then

$$\begin{aligned}\mathop{\mathbb{E}}_{\varepsilon} t(K_{2,3},S_\varepsilon) &= \mathop{\mathbb{E}}_{\varepsilon,X,Y}\Big(\mathop{\mathbb{E}}_{Z} S_\varepsilon(X,Z)S_\varepsilon(Y,Z)\Big)^3 \\ &\ge \Big(\mathop{\mathbb{E}}_{\varepsilon,X,Y}\Big(\mathop{\mathbb{E}}_{Z} S_\varepsilon(X,Z)S_\varepsilon(Y,Z)\Big)^2\Big)^{3/2} \\ &= \big(\mathop{\mathbb{E}}_{\varepsilon} t(C_4,S_\varepsilon)\big)^{3/2} = \big(8M(C_4,W)\big)^{3/2},\end{aligned}$$

where the first equality integrates out the three vertices of the part of size 3 in $K_{2,3}$, the inequality is (3.3) with $p=2$ and $q=3$, which holds since the inner expectation is nonnegative, and the last equality is (3.6). This proves the second inequality of (3.11).

Finally, since $\langle \mathbf{1}, T_{S_\varepsilon}\mathbf{1}\rangle = \mathbb{E}_{X,Y}\, S_\varepsilon(X,Y) = 1+\varepsilon t(K_2,U)$,

$$\begin{aligned}8M(C_4,W) = \mathop{\mathbb{E}}_{\varepsilon} t(C_4,S_\varepsilon) &\ge \mathop{\mathbb{E}}_{\varepsilon}\langle \mathbf{1}, T_{S_\varepsilon}\mathbf{1}\rangle^4 = \mathop{\mathbb{E}}_{\varepsilon}\big(1+\varepsilon t(K_2,U)\big)^4 \\ &= \mathop{\mathbb{E}}_{\varepsilon}\big(1+4\varepsilon t(K_2,U)+6t(K_2,U)^2+4\varepsilon t(K_2,U)^3+t(K_2,U)^4\big) \\ &= 1+6t(K_2,U)^2+t(K_2,U)^4 \ge 1,\end{aligned}$$

where the first equality is (3.6), the inequality is (3.5) with $g=\mathbf{1}$, the third equality expands the fourth power using $\varepsilon^2=1$, and the fourth equality uses $\mathbb{E}_\varepsilon\,\varepsilon = 0$. Hence $\big(8M(C_4,W)\big)^{3/2} \ge 8M(C_4,W)$, which is the last inequality of (3.11). □

3.3 Lifting to longer cycles

Let

$$V \coloneqq \left(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}\right)^{1/2},$$
$$a \coloneqq t(C_4,S_{+1}) = 16t(C_4,W), \qquad b \coloneqq t(C_4,S_{-1}) = 16t(C_4,1-W).$$

For every even $n\ge 4$, (3.8) and (3.9) give

$$2^{4n-1}M(C_n^{+3},W) \ge \mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}} V^n.$$

When $n=4$, Lemma 3.7 shows that $\mathbb{E}_{\varepsilon,\mathbf{z}}\, V^4 \ge 8M(C_4,W) = \frac{1}{2}(a+b)$, proving the three-apex inequality for the four-cycle. For larger n, note that (3.5) with $L=S_{+1}$ and $L=S_{-1}$ gives

$$\tfrac{1}{2}(a^{n/4}+b^{n/4}) \ge \tfrac{1}{2}\big(t(C_n,S_{+1})+t(C_n,S_{-1})\big) = 2^{n-1}M(C_n,W).$$

Therefore, it suffices to show that $\mathbb{E}_{\varepsilon,\mathbf{z}}\, V^n \ge \frac{1}{2}(a^{n/4}+b^{n/4})$. The following lemma shows that it suffices to split the expectation with weights ω and $2-\omega$, provided that the weight ω gives $\mathbb{E}_{\varepsilon,\mathbf{z}}\,\omega V^4 \ge \max(a,b)$.

Lemma 3.8 (Two-moment lifting)**.** *Let ζ be a random variable, let V be a bounded nonnegative function, and let $a, b \geq 0$. Suppose that a function ω satisfies*

$$0 \leq \omega(\zeta) \leq 2, \quad \mathop{\mathbb{E}}_{\zeta} \omega(\zeta) = 1, \quad \mathop{\mathbb{E}}_{\zeta} V(\zeta)^4 \geq (a+b)/2, \quad \mathop{\mathbb{E}}_{\zeta} \omega(\zeta) V(\zeta)^4 \geq \max(a, b).$$

Then for every real $q \geq 4$,

$$\mathop{\mathbb{E}}_{\zeta} V(\zeta)^q \geq \tfrac{1}{2}(a^{q/4} + b^{q/4}). \tag{3.14}$$

Proof. Without loss of generality, assume that $a \geq b$ by interchanging a and b. Since $\omega(\zeta)$ and $2 - \omega(\zeta)$ are nonnegative and have mean one, Jensen's inequality for the convex function $y \mapsto y^{q/4}$ under the probability measures with densities $\omega(\zeta)$ and $2 - \omega(\zeta)$ gives

$$\mathop{\mathbb{E}}_{\zeta} V(\zeta)^q = \tfrac{1}{2} \mathop{\mathbb{E}}_{\zeta} \omega(\zeta) V(\zeta)^q + \tfrac{1}{2} \mathop{\mathbb{E}}_{\zeta} (2 - \omega(\zeta)) V(\zeta)^q \geq \tfrac{1}{2} \big(\mathop{\mathbb{E}}_{\zeta} \omega(\zeta) V(\zeta)^4\big)^{q/4} + \tfrac{1}{2} \big(\mathop{\mathbb{E}}_{\zeta} (2 - \omega(\zeta)) V(\zeta)^4\big)^{q/4}.$$

It remains to show that the right side is at least $\frac{1}{2}(a^{q/4} + b^{q/4})$. By the fourth assumption, $\mathbb{E}_\zeta\, \omega(\zeta) V(\zeta)^4 \geq a$, so if $\mathbb{E}_\zeta (2 - \omega(\zeta)) V(\zeta)^4 \geq b$, the claim follows. Otherwise, let $\delta \coloneqq b - \mathbb{E}_\zeta (2 - \omega(\zeta)) V(\zeta)^4$, so that $0 < \delta \leq b$ and, by the third assumption,

$$\mathop{\mathbb{E}}_{\zeta} \omega(\zeta) V(\zeta)^4 = 2 \mathop{\mathbb{E}}_{\zeta} V(\zeta)^4 - \mathop{\mathbb{E}}_{\zeta} (2 - \omega(\zeta)) V(\zeta)^4 \geq (a+b) - (b - \delta) = a + \delta.$$

Then

$$\big(\mathop{\mathbb{E}}_{\zeta} \omega(\zeta) V(\zeta)^4\big)^{q/4} + \big(\mathop{\mathbb{E}}_{\zeta} (2 - \omega(\zeta)) V(\zeta)^4\big)^{q/4} \geq (a+\delta)^{q/4} + (b - \delta)^{q/4} \geq a^{q/4} + b^{q/4},$$

where the last inequality is

$$(a+\delta)^{q/4} - a^{q/4} \geq b^{q/4} - (b - \delta)^{q/4},$$

which holds because $y \mapsto y^{q/4}$ is convex and the intervals $[b - \delta, b]$ and $[a, a + \delta]$ have the same length, with $b \leq a$. □

We apply Lemma 3.8 with $\zeta = (\varepsilon, \mathbf{z})$ and with V, a, and b as above. Since $(a+b)/2 = 8M(C_4, W)$, the third condition is proven in Lemma 3.7. In the rest of this subsection, we construct a weight ω satisfying the other conditions (Proposition 3.12), and then prove the three-apex inequality.

We first prove elementary facts about kernels.

Lemma 3.9 (Signed densities)**.** *If the kernel U satisfies $|U(x,y)| \leq 1$ for all $x, y \in [0,1]$, then*

$$\begin{aligned} 0 \leq t(K_2, U)^2 \leq t(P_2, U) \leq 1, \\ 0 \leq t(C_4, U) \leq 1, \\ |t(P_3, U)| \leq t(P_2, U) t(C_4, U)^{1/4}, \end{aligned} \tag{3.15}$$

and

$$\begin{aligned} 8M(C_4, W) &= 1 + 2t(K_2, U)^2 + 4t(P_2, U) + t(C_4, U), \\ t(C_4, S_\varepsilon) &= 8M(C_4, W) + 4\varepsilon\big(t(K_2, U) + t(P_3, U)\big). \end{aligned} \tag{3.16}$$

Proof. For $x \in [0,1]$, let $d_U(x) \coloneqq \int_0^1 U(x,y)\, \mathrm{d}y$. Then

$$t(K_2, U) = \int_0^1 d_U(x)\, \mathrm{d}x, \qquad t(P_2, U) = \int_0^1 d_U(x)^2\, \mathrm{d}x,$$

Since $|U(x,y)| \le 1$, we have $|d_U(x)| \le 1$, so $t(P_2, U) \le 1$. By the Cauchy–Schwarz inequality, $t(K_2, U)^2 \le t(P_2, U)$. This proves the first line.

Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of T_U. Since $|U(x,y)| \le 1$, we have $\sum_i \lambda_i^2 = \operatorname{Tr} T_U^2 = \mathbb{E}_{X,Y}\, U(X,Y)^2 \le 1$, where X, Y are independent and uniform on $[0,1]$. Hence

$$0 \le t(C_4, U) = \sum_i \lambda_i^4 \le \Big(\sum_i \lambda_i^2\Big)^2 \le 1,$$

where the equality is (3.4), and the second inequality holds because expanding the square gives $\sum_i \lambda_i^4$ plus the nonnegative terms $\lambda_i^2 \lambda_j^2$ with $i \ne j$. This proves the second line.

The operator norm of T_U satisfies

$$\|T_U\|_{\mathrm{op}} = \max_i |\lambda_i| = \Big(\max_i \lambda_i^4\Big)^{1/4} \le \Big(\sum_i \lambda_i^4\Big)^{1/4} = t(C_4, U)^{1/4}.$$

Since $t(P_3, U) = \langle T_U \mathbf{1}, T_U(T_U \mathbf{1})\rangle$, we have $|t(P_3, U)| \le \|T_U\|_{\mathrm{op}} \|T_U \mathbf{1}\|^2 \le t(P_2, U) t(C_4, U)^{1/4}$, which proves the third line.

For the remaining two lines, (3.7) gives

$$t(C_4, S_\varepsilon) = \sum_{A \subseteq E(C_4)} \varepsilon^{|A|} t\big((C_4)_A, U\big).$$

We group the subsets A by their size, using that isolated vertices do not change the density and that $\varepsilon^2 = 1$:

$$t(C_4, S_\varepsilon) = \underbrace{1}_{|A|=0} + \underbrace{4\varepsilon t(K_2, U)}_{|A|=1} + \underbrace{2t(K_2, U)^2 + 4t(P_2, U)}_{|A|=2} + \underbrace{4\varepsilon t(P_3, U)}_{|A|=3} + \underbrace{t(C_4, U)}_{|A|=4}.$$

Among the six subsets with $|A| = 2$, two are perfect matchings, whose density is $t(K_2, U)^2$ since the two edges are disjoint, and four are pairs of adjacent edges, which form copies of P_2.

Averaging over ε, the terms with ε vanish, so

$$\mathop{\mathbb{E}}_{\varepsilon} t(C_4, S_\varepsilon) = 1 + 2t(K_2, U)^2 + 4t(P_2, U) + t(C_4, U).$$

The left side is $8M(C_4, W)$ by (3.6). This proves the fourth line.

Finally, collecting the terms with and without ε in the expansion of $t(C_4, S_\varepsilon)$ above gives

$$\begin{aligned} t(C_4, S_\varepsilon) &= \big(1 + 2t(K_2, U)^2 + 4t(P_2, U) + t(C_4, U)\big) + 4\varepsilon\big(t(K_2, U) + t(P_3, U)\big) \\ &= 8M(C_4, W) + 4\varepsilon\big(t(K_2, U) + t(P_3, U)\big), \end{aligned}$$

where the second equality is the fourth line. This proves the fifth line. □

In the proof of Proposition 3.12, we apply the following inequality with $a = 8M(C_4, W)$, $b = t(P_2, U)$, and $r = t(C_4, U)^{1/4}$, which satisfy its hypotheses by Lemma 3.9.

Lemma 3.10 (Scalar estimate)**.** *If $b \ge 0$, $0 \le r \le 1$, and $a \ge 1 + 4b + r^4$, then*

$$a^{3/2} \ge a + 4br. \tag{3.17}$$

Proof. If $r \le 1/2$, then $a \ge 1$ gives

$$a^{3/2} - a = \frac{a(a-1)}{\sqrt{a} + 1} \ge \frac{a-1}{2} \ge 2b \ge 4br.$$

If $r \geq 1/2$, the function $y^{3/2} - y$ is increasing on $[1, \infty)$, so it suffices to take $a = 1 + 4b + r^4$. For $x \coloneqq \sqrt{1 + 4b + r^4}$, we have $a = x^2$ and $4b = x^2 - 1 - r^4$, so it suffices to show that

$$a^{3/2} - a - 4br = x^3 - x^2 - r(x^2 - 1 - r^4) = x^3 - (1 + r)x^2 + r + r^5$$

is nonnegative. By grouping terms,

$$\begin{aligned} x^3 - (1 + r)x^2 + r + r^5 = (x - \tfrac{2}{3}(1 + r))^2(x + \tfrac{1}{3}(1 + r)) \\ + \tfrac{1}{27}(27r^5 - 4(1 + r)^3 + 27r). \end{aligned}$$

The first term is nonnegative since $x \geq 0$ and $r \geq 1/2$, so it is enough to check that the last polynomial in r is nonnegative. With $t = r - 1/2 \geq 0$, this polynomial equals

$$27t^5 + \frac{135}{2}t^4 + \frac{127}{2}t^3 + \frac{63}{4}t^2 + \frac{135}{16}t + \frac{27}{32},$$

whose coefficients are positive. □

We construct the weight ω using the polynomial

$$\mathcal{P} \coloneqq U(z_1, z_2) + U(z_1, z_3) + U(z_2, z_3) - U(z_1, z_2)U(z_1, z_3)U(z_2, z_3). \tag{3.18}$$

The polynomial $x + y + z - xyz$ is multiaffine on $[-1, 1]^3$ and takes the values ± 2 at the corners, so

$$|\mathcal{P}| \leq 2. \tag{3.19}$$

Its mean is

$$\mathop{\mathbb{E}}_{\varepsilon, \mathbf{z}} \mathcal{P} = 3t(K_2, U) - t(K_3, U). \tag{3.20}$$

Lemma 3.11 (Weighted certificates)**.** *For every graphon W, with $U = 2W - 1$,*

$$\begin{aligned} &\mathop{\mathbb{E}}_{\varepsilon, \mathbf{z}}(3 + \varepsilon - \mathcal{P})\big(2t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon) - t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)^2\big) \\ &\qquad \geq 24M(C_4, W) + 13t(K_2, U) + t(K_3, U) \\ &\qquad\qquad + 12t(K_2, U)^2 - 4t(K_2, U)t(K_3, U) + 12t(P_3, U), \qquad (3.21) \\ &\mathop{\mathbb{E}}_{\varepsilon, \mathbf{z}}(6 + \varepsilon + \mathcal{P})\big(2t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon) - t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)^2\big) \\ &\qquad \geq 48M(C_4, W) + 19t(K_2, U) + 23t(K_3, U) \\ &\qquad\qquad - 12t(K_2, U)^2 + 4t(K_2, U)t(K_3, U) + 24t(P_3, U). \qquad (3.22) \end{aligned}$$

Proof. These inequalities are proven by flag algebra certificates, where the specification of these problems is described in Appendix B.1. □

Proposition 3.12 (Weighted comparison)**.** *There is a function ω of $(\varepsilon, \mathbf{z})$ such that $0 \leq \omega \leq 2$, $\mathbb{E}_{\varepsilon, \mathbf{z}}\, \omega = 1$, and*

$$\mathop{\mathbb{E}}_{\varepsilon, \mathbf{z}} \omega \left(\frac{t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)} \right)^2 \geq 16 \max\{t(C_4, W), t(C_4, 1 - W)\}. \tag{3.23}$$

Proof. Without loss of generality, assume that $t(K_2, U) \geq 0$ by interchanging W and $1 - W$; a weight for $1 - W$ gives a weight for W after replacing ε by $-\varepsilon$. We abbreviate $\mathbb{E} = \mathbb{E}_{\varepsilon, \mathbf{z}}$, so that the left side of (3.23) is $\mathbb{E}\, \omega V^4$. By the fifth line of Lemma 3.9 with $\varepsilon = +1$ and $\varepsilon = -1$,

$$\begin{aligned} 16t(C_4, W) &= 8M(C_4, W) + 4\big(t(K_2, U) + t(P_3, U)\big), \\ 16t(C_4, 1 - W) &= 8M(C_4, W) - 4\big(t(K_2, U) + t(P_3, U)\big). \end{aligned}$$

Suppose first that $t(K_2,U)+t(P_3,U)\le 0$, so that $16t(C_4,1-W)\ge 16t(C_4,W)$. If we take $\omega=1$, which satisfies $0\le\omega\le 2$ and $\mathbb{E}\,\omega=1$, then

$$\begin{aligned}\mathbb{E}\,V^4 &\ge \big(8M(C_4,W)\big)^{3/2} \ge 8M(C_4,W)+4t(P_2,U)t(C_4,U)^{1/4}\\ &\ge 8M(C_4,W)-4\big(t(K_2,U)+t(P_3,U)\big)=16t(C_4,1-W),\end{aligned}$$

where the first inequality is (3.11); the second is Lemma 3.10 with $a=8M(C_4,W)$, $b=t(P_2,U)$, and $r=t(C_4,U)^{1/4}$, whose hypotheses are proven in the first, second, and fourth lines of Lemma 3.9; the third uses $t(K_2,U)\ge 0$ and the third line of Lemma 3.9; and the equality is the identity for $16t(C_4,1-W)$ shown above.

Suppose now that $t(K_2,U)+t(P_3,U)>0$, so that $16t(C_4,W)>16t(C_4,1-W)$. We first show that

$$2^{12}M(K_{1,2,3},W)-2^6M(K_{2,3},W)-8M(C_4,W)-4t(P_3,U)-1\ge 0. \tag{3.24}$$

Since $\langle\mathbf{1},T_{S_{-1}}\mathbf{1}\rangle=1-t(K_2,U)$, (3.5) with $L=S_{-1}$ and $g=\mathbf{1}$ gives $16t(C_4,1-W)=t(C_4,S_{-1})\ge(1-t(K_2,U))^4$. Together with the identity for $16t(C_4,1-W)$ shown above, this gives

$$\begin{aligned}8M(C_4,W)-4t(P_3,U)&=16t(C_4,1-W)+4t(K_2,U)\ge(1-t(K_2,U))^4+4t(K_2,U)\\ &=1+t(K_2,U)^2\big((t(K_2,U)-2)^2+2\big)\ge 1.\end{aligned}$$

By (3.11), $2^{11}M(K_{1,2,3},W)-2^5M(K_{2,3},W)\ge 8M(C_4,W)$, so the left side of (3.24) is at least $2\cdot 8M(C_4,W)-8M(C_4,W)-4t(P_3,U)-1=8M(C_4,W)-4t(P_3,U)-1\ge 0$. This proves (3.24).

Next, we rewrite the right sides of (3.21) and (3.22). Substituting $t(K_3,U)=3t(K_2,U)-\mathbb{E}\,\mathcal{P}$ from (3.20) and $8M(C_4,W)+4t(K_2,U)+4t(P_3,U)=16t(C_4,W)$ shown above, the right side of (3.21) becomes

$$48t(C_4,W)+4t(K_2,U)-\mathbb{E}\,\mathcal{P}+4t(K_2,U)\,\mathbb{E}\,\mathcal{P},$$

and the right side of (3.22) becomes

$$96t(C_4,W)+64t(K_2,U)-23\,\mathbb{E}\,\mathcal{P}-4t(K_2,U)\,\mathbb{E}\,\mathcal{P}.$$

We now define the weight by

$$\omega\coloneqq\begin{cases}(3+2\,\mathbb{E}\,\mathcal{P}+\varepsilon-\mathcal{P})/(3+\mathbb{E}\,\mathcal{P}), & \mathbb{E}\,\mathcal{P}\ge 0,\\ (6-2\,\mathbb{E}\,\mathcal{P}+\varepsilon+\mathcal{P})/(6-\mathbb{E}\,\mathcal{P}), & \mathbb{E}\,\mathcal{P}<0.\end{cases} \tag{3.25}$$

If $\mathbb{E}\,\mathcal{P}\ge 0$, then $\omega=(3+2\,\mathbb{E}\,\mathcal{P}+\varepsilon-\mathcal{P})/(3+\mathbb{E}\,\mathcal{P})$. Since $\varepsilon\in\{\pm1\}$ and $|\mathcal{P}|\le 2$ by (3.19), its numerator satisfies

$$0\le 2\,\mathbb{E}\,\mathcal{P}\le 3+2\,\mathbb{E}\,\mathcal{P}+\varepsilon-\mathcal{P}\le 6+2\,\mathbb{E}\,\mathcal{P}=2(3+\mathbb{E}\,\mathcal{P}),$$

so $0\le\omega\le 2$. Since $\mathbb{E}\,\varepsilon=0$, the mean of the numerator is $3+\mathbb{E}\,\mathcal{P}$, so $\mathbb{E}\,\omega=1$. In this case, we prove that $\mathbb{E}\,\omega V^4\ge 16t(C_4,W)$. We have

$$\begin{aligned}&(3+\mathbb{E}\,\mathcal{P})\big(\mathbb{E}\,\omega V^4-16t(C_4,W)\big)\\ &=\mathbb{E}(3+2\,\mathbb{E}\,\mathcal{P}+\varepsilon-\mathcal{P})V^4-16(3+\mathbb{E}\,\mathcal{P})t(C_4,W)\\ &\ge\mathbb{E}(3+\varepsilon-\mathcal{P})\big(2t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)-t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2\big)\\ &\qquad+2\,\mathbb{E}\,\mathcal{P}\big(2^{11}M(K_{1,2,3},W)-2^5M(K_{2,3},W)\big)-16(3+\mathbb{E}\,\mathcal{P})t(C_4,W)\\ &\ge 4t(K_2,U)-\mathbb{E}\,\mathcal{P}+4t(K_2,U)\,\mathbb{E}\,\mathcal{P}\\ &\qquad+2\,\mathbb{E}\,\mathcal{P}\big(2^{11}M(K_{1,2,3},W)-2^5M(K_{2,3},W)\big)-16\,\mathbb{E}\,\mathcal{P}\,t(C_4,W)\\ &=4t(K_2,U)+\mathbb{E}\,\mathcal{P}\big(2^{12}M(K_{1,2,3},W)-2^6M(K_{2,3},W)-8M(C_4,W)-4t(P_3,U)-1\big)\\ &\ge 0,\end{aligned}$$

where the first inequality is (3.12), which holds since $3+\varepsilon-\mathcal{P}\ge 0$ and $2\,\mathbb{E}\,\mathcal{P}\ge 0$, together with (3.13); the second is (3.21) in the form above; the second equality uses $16t(C_4,W) = 8M(C_4,W)+4t(K_2,U)+4t(P_3,U)$; and the last inequality uses $t(K_2,U)\ge 0$, $\mathbb{E}\,\mathcal{P}\ge 0$, and (3.24). Since $3+\mathbb{E}\,\mathcal{P}>0$, this proves (3.23).

If $\mathbb{E}\,\mathcal{P}<0$, then $\omega = (6-2\,\mathbb{E}\,\mathcal{P}+\varepsilon+\mathcal{P})/(6-\mathbb{E}\,\mathcal{P})$. Since $\varepsilon\in\{\pm1\}$ and $|\mathcal{P}|\le 2$ by (3.19), its numerator satisfies

$$0<3-2\,\mathbb{E}\,\mathcal{P}\le 6-2\,\mathbb{E}\,\mathcal{P}+\varepsilon+\mathcal{P}\le 9-2\,\mathbb{E}\,\mathcal{P}<2(6-\mathbb{E}\,\mathcal{P}),$$

so $0\le\omega\le 2$. Since $\mathbb{E}\,\varepsilon=0$, the mean of the numerator is $6-\mathbb{E}\,\mathcal{P}$, so $\mathbb{E}\,\omega=1$. In this case, we prove that $\mathbb{E}\,\omega V^4\ge 16t(C_4,W)$. We have

$$\begin{aligned}
&(6-\mathbb{E}\,\mathcal{P})\big(\mathbb{E}\,\omega V^4-16t(C_4,W)\big)\\
&=\mathbb{E}(6-2\,\mathbb{E}\,\mathcal{P}+\varepsilon+\mathcal{P})V^4-16(6-\mathbb{E}\,\mathcal{P})t(C_4,W)\\
&\ge\mathbb{E}(6+\varepsilon+\mathcal{P})\big(2t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)-t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)^2\big)\\
&\qquad-2\,\mathbb{E}\,\mathcal{P}\big(2^{11}M(K_{1,2,3},W)-2^5M(K_{2,3},W)\big)-16(6-\mathbb{E}\,\mathcal{P})t(C_4,W)\\
&\ge 64t(K_2,U)-23\,\mathbb{E}\,\mathcal{P}-4t(K_2,U)\,\mathbb{E}\,\mathcal{P}\\
&\qquad-2\,\mathbb{E}\,\mathcal{P}\big(2^{11}M(K_{1,2,3},W)-2^5M(K_{2,3},W)\big)+16\,\mathbb{E}\,\mathcal{P}\,t(C_4,W)\\
&=64t(K_2,U)-24\,\mathbb{E}\,\mathcal{P}\\
&\qquad-\mathbb{E}\,\mathcal{P}\big(2^{12}M(K_{1,2,3},W)-2^6M(K_{2,3},W)-8M(C_4,W)-4t(P_3,U)-1\big)\\
&\ge 0,
\end{aligned}$$

where the first inequality is (3.12), which holds since $6+\varepsilon+\mathcal{P}\ge 0$ and $-2\,\mathbb{E}\,\mathcal{P}>0$, together with (3.13); the second is (3.22) in the form above; the second equality uses $16t(C_4,W) = 8M(C_4,W)+4t(K_2,U)+4t(P_3,U)$; and the last inequality uses $t(K_2,U)\ge 0$, $\mathbb{E}\,\mathcal{P}<0$, and (3.24). Since $6-\mathbb{E}\,\mathcal{P}>0$, this proves (3.23). □

Proposition 3.13 (Three apices)**.** *For every even $n\ge 4$,*

$$M(C_n^{+3},W)\ge 2^{-3n}M(C_n,W). \tag{3.26}$$

Proof. Apply Lemma 3.8 with $\zeta=(\varepsilon,\mathbf{z})$, $V=\big(t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)/t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)\big)^{1/2}$, $a=16t(C_4,W)$, and $b=16t(C_4,1-W)$, where the third assumption is proven in Lemma 3.7 and the other assumptions are proven in Proposition 3.12. Then

$$\begin{aligned}
2^{4n-1}M(C_n^{+3},W)=\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}t_{\mathbf{z}}(C_n^{+3},S_\varepsilon)&\ge\mathop{\mathbb{E}}_{\varepsilon,\mathbf{z}}\Big(\frac{t_{\mathbf{z}}(P_2^{+3},S_\varepsilon)}{t_{\mathbf{z}}(K_1^{+3},S_\varepsilon)}\Big)^{n/2}\\
&\ge 2^{n-1}\big(t(C_4,W)^{n/4}+t(C_4,1-W)^{n/4}\big)\\
&\ge 2^{n-1}\big(t(C_n,W)+t(C_n,1-W)\big)=2^{n-1}M(C_n,W),
\end{aligned}$$

where the first equality is (3.8) with $e(C_n^{+3})=4n$, the first inequality is (3.9), the second is (3.14) with $q=n$, and the third is $t(C_n,L)\le t(C_4,L)^{n/4}$ from (3.5) with $L=W$ and $L=1-W$. □

Proof of Theorem 1.4. For $k\ge 3$, Proposition 3.13 and (3.2) give $M(C_n^{+k},W)\ge 2^{-nk}M(C_n,W)$ for every step graphon W, and hence for every graphon W by Lemma 3.2. Since C_n is common, this gives (1.6), which implies (1.5). For $k=1$ and $k=2$, (1.5) is known [15], [5, Theorem 4.3]. □

4 Remark on the Lean verification

Theorems 1.2, 1.3, and 1.4 have been formalized in the Lean 4 proof assistant [2], on top of the mathlib library [16]. The formalization also proves the cases $k = 1$ and $k = 2$ of Theorem 1.4, which this paper cites from the literature, and shows that equality in (1.5) holds only if $W = 1/2$ almost everywhere. The formal proof differs from the one given here in two ways.

First, instead of approximating graphons by finite graphs in Section 2 and by step graphons in Section 3, both parts prove the inequalities for weighted finite hosts, in which nonnegative vertex weights sum to one and the kernel may be nonzero on the diagonal. These hosts represent step graphons on finite measurable partitions, and continuity of the densities under L^1 approximation transfers the inequalities to graphons on an arbitrary probability space. In Section 3, the matrix T_L is replaced by the matrix with entries $\sqrt{w_i w_j}\, L(i,j)$, where w_i are the vertex weights.

Second, a weighted host has no uniformly random homomorphism, so the tree proof uses, instead of the Shannon entropy of a law p, its relative entropy $\sum_{a:p(a)>0} p(a)\log(q(a)/p(a))$ against the weight q of a vertex map in the homomorphism density. Gibbs' inequality, which bounds this by $\log \sum_a q(a)$, replaces the support bound (2.2); for a graph with uniform weights the two bounds coincide. In the proof of (2.11), Gibbs' inequality is applied to the law of $(Z_{1:k}, X)$ against the law in which the pages are conditionally independent given X, in place of the subadditivity of mutual information (2.5).

A Reduction to step graphons

Proof of Lemma 3.2. We first show that every graph F and all symmetric measurable $L, L' : [0,1]^2 \to [0,1]$ satisfy

$$|t(F,L) - t(F,L')| \le e(F)\,\|L - L'\|_1. \tag{A.1}$$

Let $e_1, \dots, e_m$ be the edges of F, and write $L(e) \coloneqq L(x_u, x_v)$ for $e = uv$. Then

$$\prod_{l=1}^{m} L(e_l) - \prod_{l=1}^{m} L'(e_l) = \sum_{l=1}^{m} \left(\prod_{l'<l} L'(e_{l'}) \right) (L(e_l) - L'(e_l)) \left(\prod_{l'>l} L(e_{l'}) \right).$$

All factors in the products lie in $[0,1]$, so the lth term is at most $|L(e_l) - L'(e_l)|$ in absolute value, whose integral over $[0,1]^{V(F)}$ is $\|L - L'\|_1$. Integrating gives (A.1).

Let W be a graphon. For $j \ge 0$, let $d \coloneqq 2^j$, and let W_j be the step graphon equal to $d^2 \int_{I_\alpha \times I_\beta} W$ on each $I_\alpha \times I_\beta$. Then W_j is the conditional expectation of W given the σ-algebra generated by these squares. These σ-algebras increase with j and generate the Borel σ-algebra of $[0,1]^2$, so $W_j \to W$ in L^1 by the martingale convergence theorem. Since also $\|(1 - W_j) - (1 - W)\|_1 = \|W_j - W\|_1$, (A.1) gives $M(F, W_j) \to M(F, W)$ for every graph F. By assumption, (3.2) holds for every W_j, and letting $j \to \infty$ on both sides gives (3.2) for W. □

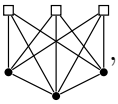
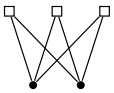
B Exact flag algebra certificates

We specify the three certificate identities used in Lemmas 3.6 and 3.11. The presentation uses signed homomorphism densities rather than induced colored densities; the positivity mechanism is the rooted quadratic-form principle of flag algebras [12]. All three certificates use the common feature schema described below.

B.1 Targets in the signed graph algebra

For a graph G, write $[G] \coloneqq t(G, U)$. The certificates show that Q_3, Q_-, and Q_+ are nonnegative, where $Q_3 \coloneqq 2^{10} M(K_{1,2,3}, W) - 2^5 M(K_{2,3}, W)$, and Q_- and Q_+ are the left sides minus the right sides of (3.21) and (3.22). By (3.6) and (3.7), every Ramsey multiplicity is a linear combination of the $[G]$, and a product $[G][H]$ is $[G \sqcup H]$, where $G \sqcup H$ is the disjoint union. Hence Q_3 and the right sides of (3.21) and (3.22) are linear combinations of $[G]$ over graphs G with at most six vertices.

It remains to write the left sides of (3.21) and (3.22) in this way. We draw the three apices as squares ▫ and the other vertices as dots, and let

$E_3 \coloneqq$  , $\qquad B_3 \coloneqq$ 

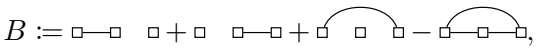
be the copies of $K_{1,2,3} = P_2^{+3}$ and $K_{2,3} = \overline{K_2}^{+3}$ whose part of size 3 consists of the apices. For such a graph G and $\eta \in \{0, 1\}$, let

$$\mathsf{P}_\eta(G) \coloneqq \sum_{\substack{A \subseteq E(G) \\ |A| \equiv \eta \pmod 2}} [G_A], \qquad F_\eta \coloneqq 2\mathsf{P}_\eta(E_3) - \mathsf{P}_\eta(B_3),$$

where G_A has vertex set $V(G)$ and edge set A, with the apices kept as squares. Placing the apices at $\mathbf{z}$ and expanding as in (3.7), the factor $\varepsilon^{|A|}$ of G_A has mean 1 when $|A|$ is even and 0 otherwise, and $\varepsilon \cdot \varepsilon^{|A|}$ has mean 1 when $|A|$ is odd and 0 otherwise. Hence

$$\mathop{\mathbb{E}}_{\varepsilon, \mathbf{z}} \varepsilon^\eta \big(2t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon) - t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)^2\big) = F_\eta.$$

Multiplying by $U(z_i, z_j)$ before integrating over $\mathbf{z}$ adds an edge between the apices i and j. Hence, with

$B \coloneqq$  ,

we have

$$\mathop{\mathbb{E}}_{\varepsilon, \mathbf{z}} \mathcal{P}\big(2t_{\mathbf{z}}(P_2^{+3}, S_\varepsilon) - t_{\mathbf{z}}(K_1^{+3}, S_\varepsilon)^2\big) = BF_0,$$

where BF_0 adds the edges of each graph in B to each graph in F_0, on the same apices. Therefore the left sides of (3.21) and (3.22) are $3F_0 + F_1 - BF_0$ and $6F_0 + F_1 + BF_0$, respectively.

B.2 Rooted quadratic forms

Choose r roots and ℓ further vertices. A feature graph A has no root–root edges. Its rooted density is

$$\phi_A(\mathbf{z}) \coloneqq \int \prod_{ij \in E(A)} U(x_i, x_j)\, \mathrm{d}\mathbf{x},$$

where the root coordinates are fixed at $\mathbf{z}$ and the other coordinates are integrated independently. For a root type $t \subseteq \binom{\{0,\dots,r-1\}}{2}$, let

$$p_t(\mathbf{z}) \coloneqq 2^{-\binom{r}{2}} \prod_{e \in t} (1 + U_e) \prod_{e \notin t} (1 - U_e).$$

For a feature vector ϕ and a positive semidefinite matrix Q, the expression

$$\int p_t(\mathbf{z}) \phi(\mathbf{z})^\mathsf{T} Q \phi(\mathbf{z})\, \mathrm{d}\mathbf{z} \tag{B.1}$$

is nonnegative.

Each certificate consists of a denominator $\mathscr{D}$ and nineteen integer matrices $A_1, \ldots, A_{19}$, and the quadratic-form matrices are $A_j/\mathscr{D}$. The rows and columns of A_j are indexed by the features of the jth block, which has r roots, ℓ free vertices, and root type t_j. The certificates for Q_3, Q_-, and Q_+, together with the blocks and their features, are available at `https://github.com/mekty2012/Apex-Commonality`.

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